Byte Blast — Game Design Handout

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Overview

Pitch: A block-placement puzzle where every row is a byte, and you clear lines by building the right binary number.

Byte Blast takes the proven Block Blast / Woodoku loop and makes the 8-column grid mean something. Each column is a bit with a fixed place value (128 down to 1). A row clears only when its filled cells add up to the row's target number. Players learn binary place values by playing, not by drilling.

Item	Detail
Genre	Casual block puzzle, educational
Learning goal	Read and write 8-bit binary numbers; later hex and ASCII
Primary audience	Ages 11–18, intro CS and computing students
Secondary audience	Teachers, adult self-learners, IT certification candidates
Session length	3–8 minutes per run
Platforms	Web first (school Chromebooks), then iOS and Android
Business model	Free web version; mobile freemium with optional ads removal

Design pillars:

1. **Fun first.** It must be worth playing even if you ignore the learning.
2. **The mechanic teaches.** No quiz pop-ups; place value is learned by placing blocks.
3. **Instant feedback.** Every move shows the row's current value live.
4. **Short and replayable.** Quick runs, daily goals, easy to pick up in class.

Core loop and rules

The player places pieces on an 8×8 grid; a row clears when its filled cells equal the row's target number.

The board

- 8 columns, each labeled with its place value along the top: 128 · 64 · 32 · 16 · 8 · 4 · 2 · 1.
- 8 rows. Each row shows a **target** on the right (e.g. 37) and its **live value** as cells fill (e.g. 32 + 4 = 36).
- A filled cell is a 1; an empty cell is a 0. The row's binary string displays under the live value (00100100).

The turn

1. Three pieces appear in the tray (classic polyomino shapes: singles, dominoes, L-shapes, lines).
2. The player drags each piece onto the grid. Pieces can't overlap filled cells.
3. After each placement, every touched row updates its live value.
4. When all three pieces are placed, three new ones appear.

Clearing

- **Exact match:** a row whose value equals its target clears, scores, and gets a new target.
- **Full byte (255):** a completely filled row always clears, but scores less than an exact match. This keeps the classic Block Blast feel and prevents dead boards.
- **Columns** clear when fully filled, as in the original game; no target applies to columns.
- Clearing several rows with one placement triggers a combo.

Overflow and game over

- A row whose value passes its target is **overflowed**: it turns amber and can only clear by reaching 255. This teaches that every bit adds value.
- The run ends when none of the three tray pieces fits anywhere on the grid.

Worked example: target 37 = 32 + 4 + 1 → 00100101. The player needs cells in the 32, 4 and 1 columns filled, and nothing else in that row.

Learning design

Difficulty grows by unlocking bits and changing what the target looks like, so players move from counting to reading binary fluently.

Stage	What changes	What the player learns
1. Nibble (tutorial)	Only the 4 rightmost columns are active (8·4·2·1); targets 1–15	Each column doubles; small numbers are sums of 8, 4, 2, 1
2. Full byte	All 8 columns active; targets 1–255	Place values up to 128; reading a full byte
3. Reverse	Target shown in binary; live value shown in decimal	Converting binary → decimal
4. Hidden helpers	Column labels fade out after 30 seconds	Recalling place values from memory
5. Speed	Targets refresh on a timer	Fluency and mental conversion speed

Learning supports built into play:

- **Live sum bar:** each row shows its running total, so the math is always visible.
- **Hint glow:** after 10 idle seconds, the columns still needed for a target softly glow. Hints cost points, never progress.
- **Bit facts:** short one-line tips between runs ("Odd numbers always use the 1 column").
- **Mastery tracking:** the game records which targets a player solves slowly and serves them more often.

Learning outcomes a teacher can check: convert any 0–255 number to binary and back; explain why each bit doubles; recognize powers of two on sight.

Game modes

Launch with Classic and Puzzle; add the others once the core loop tests well.

Mode	How it plays	Teaches	Release
Classic	Endless run, play until no piece fits; chase a high score	Fluency through repetition	MVP
Puzzle levels	Hand-made boards with fixed pieces and 1–3 targets; solve in limited moves	Planning which bits to set	MVP

Mode	How it plays	Teaches	Release
Daily Byte	One shared puzzle per day, same for everyone, with a streak counter	Habit and steady practice	v1.1
Hex mode	Targets shown in hexadecimal (e.g. 0x2F); grid splits into two nibbles	Hex ↔ binary, why one hex digit = 4 bits	v1.2
ASCII mode	Each cleared row reveals a letter (01001000 = H); a run spells a hidden word	Text encoding	v1.2
Classroom	Teacher shares a code; students play the same seed; teacher sees a results board	Group practice, assessment	v1.3

Progression, scoring and rewards

Scoring rewards exact binary answers far more than brute-force filling, so learning is always the best strategy.

Event	Points
Each cell placed	1
Full row (255) cleared	20
Exact target row cleared	50 + the target value ÷ 5
Column cleared	20
Combo (2+ lines at once)	Total × number of lines
Streak (exact clears on consecutive turns)	+10% per turn, up to +50%
Hint used	−15

Long-term progression:

- **XP and levels:** every run earns XP; levels unlock new modes and stages from the learning table.
- **Bit badges:** achievements such as "Powers of Two" (clear 1, 2, 4 … 128), "Odd One Out" (10 odd targets), "Full House" (clear 255 as an exact target).

- **Cosmetics:** block skins and themes (circuit board, retro terminal, neon), earned through play.
- **Personal bests:** high score per mode and fastest Daily Byte time.

UI/UX and visual style

The screen is one portrait column: score on top, the board with its labels in the middle, the piece tray at the bottom.

Screen layout (top to bottom)

1. Header: score, best score, pause.
2. Place-value row: 128 · 64 · 32 · 16 · 8 · 4 · 2 · 1 above the columns.
3. Board: 8×8 grid. Right of each row: target, live value, and the row's binary string in small monospace.
4. Piece tray: three pieces, large touch targets.

Feedback

- Ghost preview while dragging shows the row's new value before you drop.
- Row states by color: neutral (under target), green pulse (exact match), amber (overflow).
- Clear animation: the row's bits light up left to right, then the number flies to the score.
- Short sound per bit on clear, rising in pitch with place value.

Art direction: clean and bright, inspired by circuit boards and retro terminals. Rounded blocks, a dark navy board, and a high-contrast accent per state. The look must be clearly original, not a reskin of Block Blast.

Accessibility

- Row states never rely on color alone: icons (✓ exact, ▲ overflow) appear beside the value.
- Colorblind-safe palette and a high-contrast theme.
- Tap-to-select and tap-to-place as an alternative to dragging.
- Readable at 360 px wide; minimum 44 px touch targets.
- Sound off by default in Classroom mode.

Technical plan

Build one web codebase first and wrap it for mobile, so the school version and the app version stay identical.

Layer	Choice	Why
Game client	TypeScript + HTML Canvas (or Phaser)	Runs on Chromebooks, no install
Mobile	Capacitor wrapper around the web build	One codebase for web, iOS, Android
Save data	Local storage on device; optional account later	No login needed for students
Classroom backend	Small serverless API + database (v1.3)	Shared seeds and result boards
Analytics	Privacy-first, no personal data from minors	School and COPPA requirements

Key algorithms

- **Row value:** each row is stored as an 8-bit integer; placing a cell at column c sets bit (7 − c). The live value is just that integer, so checks are instant.
- **Target generation:** pick targets the current tray could plausibly reach. Early stages weight toward numbers with few 1-bits (e.g. 4, 8, 12); later stages use the full 1–255 range.
- **Solvability guard:** before dealing a tray, simulate placements to confirm at least one piece fits. Re-roll unfair trays, capped at a few tries to keep things random.
- **Seeded randomness:** a seed drives pieces and targets, which makes Daily Byte and Classroom runs identical for everyone.

Privacy: the game is aimed at minors, so collect no personal data by default and follow COPPA (US) and GDPR-K (EU) rules before adding accounts or ads.

Monetization and distribution

Win schools with a free, ad-free web version, then earn from mobile players and paid classroom features.

Channel	Offer	Revenue
Web (schools)	Free, no ads, no login	None directly; drives awareness
Mobile app	Free with optional rewarded ads (extra tray re-roll, hint)	Ad revenue
Mobile app	One-time "Remove ads" purchase	In-app purchase
Mobile app	Cosmetic theme packs	In-app purchase
Classroom Pro	Teacher dashboard, class reports, custom puzzle sets	School or district subscription

Distribution plan:

1. **Teachers first:** share on computing-teacher communities and forums, and list the game on education resource sites.
2. **Lesson kit:** a free one-page lesson plan and worksheet that use the game in a 20-minute class.
3. **Shareable moments:** Daily Byte results share as a simple emoji grid, like Wordle.
4. **App stores:** launch mobile once web retention looks healthy, with store keywords such as binary game, learn binary, and coding puzzle.

Competition and differentiation

No existing game combines block-placement puzzles with binary, so Byte Blast competes with drills and logic puzzles rather than a direct rival.

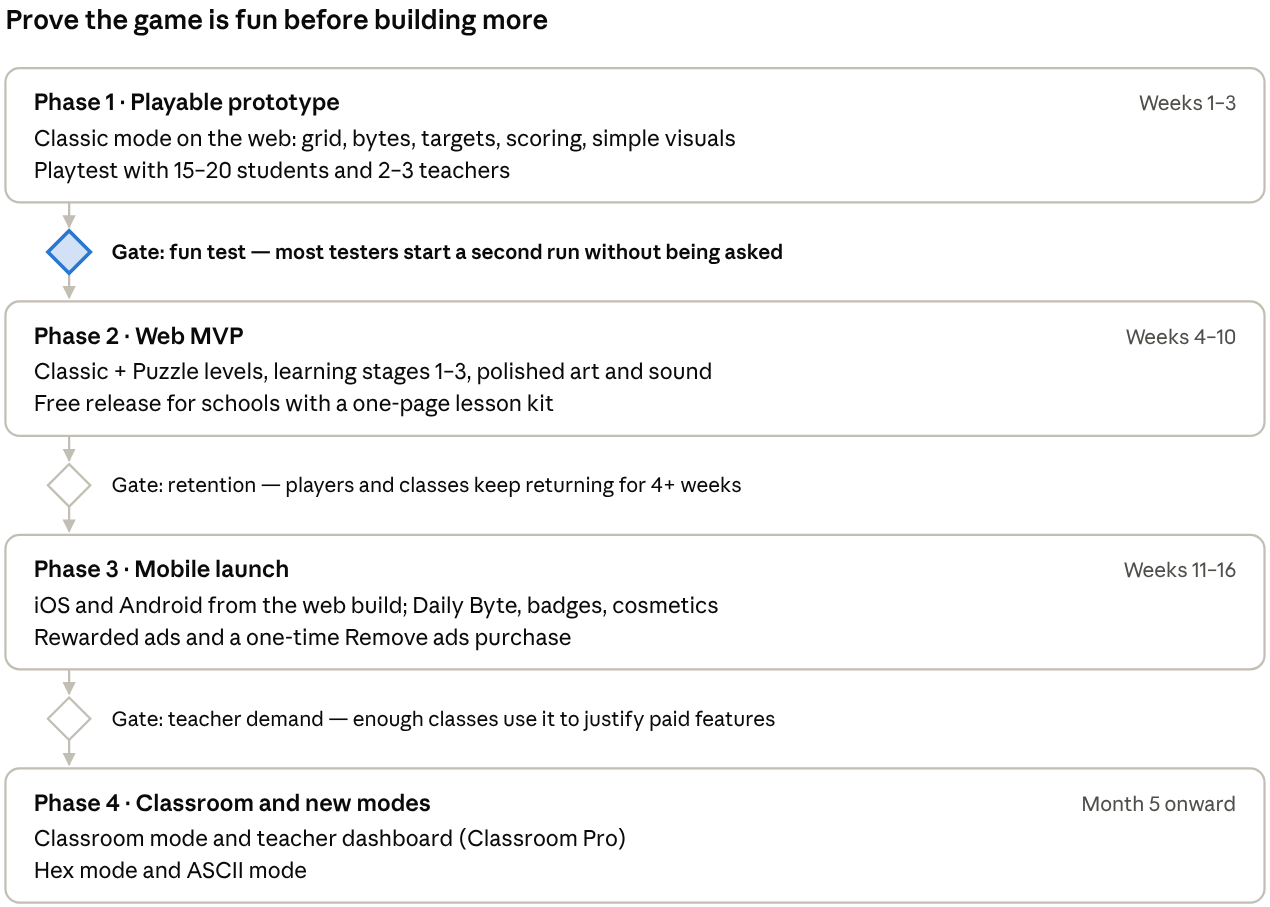
Product	What it does	Gap Byte Blast fills
Cisco Binary Game	Rows of bits rise from the bottom; click bits to build numbers before the stack fills	A conversion drill, not a puzzle; little strategy or replay depth
0h h1	Binary-sudoku logic grids (Takuzu / Binairo)	Uses two colors, but teaches no place values
Block Blast / Woodoku	Hugely popular block placement	Pure entertainment; teaches nothing

Product	What it does	Gap Byte Blast fills
Classroom quiz tools	Binary conversion quizzes and worksheets	Feels like homework; no game loop

What makes Byte Blast different: a top-chart mechanic players already love; learning built into every move; and a classroom mode that spreads through teachers.

Roadmap, metrics and risks

Ship a playable prototype in about 3 weeks, and scale up only after it passes a fun test with real students.



Byte Blast roadmap · 4 phases, 3 gates

Durations are estimates for one or two developers. Each gate decides whether to continue, adjust, or stop.

Success metrics

Metric	How to measure	What good looks like
Fun	Watch playtests; count unprompted second runs	Most testers replay without being asked
Learning	Short binary quiz before and after 3 sessions	Faster, more accurate 0–255 conversions
Retention	Day-1 and day-7 return rates	At or above casual puzzle benchmarks
Reach	Active classes and teacher sign-ups per month	Steady month-over-month growth
Revenue	Ad and purchase revenue per mobile player	Covers hosting and development costs

Risks and mitigations

Risk	Mitigation
Feels like homework	Full-byte clears always work; targets ramp slowly; no quiz pop-ups
Too hard for beginners	Nibble tutorial stage, live sums, hint glow
Seen as a Block Blast copy	Original name, art, and visual style; reuse only the general mechanic
Copycats after launch	Ship fast; build a teacher community and Classroom features others lack
Child privacy rules	No personal data by default; COPPA and GDPR-K review before ads or accounts
Ads in schools	Keep the web version ad-free